Motor Control & Networks

1. In a stepper motor, what is the maximum torque the motor can withstand without moving from its position with the stator energized?

a. Detent torque
b. Holding torque
c. Pull-in torque
d. Pull-out torque

2. In a stepper motor, what is the maximum torque the motor can withstand without moving from its position with the stator de-energized?

a. Holding torque
b. Pull-in torque
c. Detent torque
d. Running torque

3. When the system is more capacitive, how is the phase of the current related to the voltage, and what power factor does the system have?

a. The current is in phase with the voltage, and the system has a unity power factor.
b. The current is out of phase with the voltage, and the system has a leading power factor.
c. The current is out of phase with the voltage, and the system has a lagging power factor.
d. The current is 180 degrees out of phase with the voltage, and the system has a zero power factor.

4. Which type of motor must have a position feedback sensor?

a. Servo motors
b. Stepper motors
c. Squirrel-cage induction motors
d. Shaded-pole motors

9. In the control system block diagram, which lines represent a divergent control path?

a. 1 and 2
b. 3 and 8
c. 5, 6, and 7
d. 7 and 9

10. In the control system block diagram, which line represents a feedback control path?

a. Line 3
b. Line 4
c. Line 7
d. Line 9

11. In the motor control troubleshooting steps, where does the missing "Correct the problem" step go?

a. Between DEFINE THE PROBLEM and DECIDE WHAT NEEDS TO BE TESTED

b. Between DECIDE WHAT NEEDS TO BE TESTED and DETERMINE THE CAUSE OF THE FAILURE

c. Between DECIDE WHAT TYPE OF TEST TO PERFORM and VERIFY CORRECT OPERATION

d. After VERIFY CORRECT OPERATION

Hazardous Locations

15. To minimize the effects of "pressure piling," how far apart should seals generally be inserted in long conduit runs?

a. Not more than 10-25 ft apart

b. Not more than 25-50 ft apart

c. Not more than 50-100 ft apart, depending on conduit size

d. Not more than 100-150 ft apart, regardless of conduit size

16. Where ventilation is not provided in a minor repair garage, what floor area is considered Division 2, Zone 2?

a. The area up to 12 in. above the floor, extending 10 ft horizontally in all directions

b. The area up to 18 in. above the floor, extending 20 ft horizontally in all directions

c. The area up to 24 in. above the floor, extending 5 ft horizontally in all directions

d. The entire floor area up to 36 in. above the floor

17. What is vapor density?

a. The ratio of a liquid's weight to the weight of an equal volume of water

b. The amount of vapor present in air at the saturation point

c. The weight of a volume of pure vapor or gas compared with the weight of an equal volume of dry air at the same temperature and pressure

d. The minimum concentration of vapor required for ignition

Security & Access Control

18. What is a partition in a security system?

a. A portion of the security system that operates under a different program than the rest of the system

b. A supervised communication circuit between the control panel and central station

c. A device that physically separates two security zones

d. A section of a system that always follows the same program as every other section

19. What type of security system circuit uses devices connected in series?

a. Closed-loop circuit

b. Open-loop circuit

c. Addressable-loop circuit

d. Parallel supervisory circuit

20. An open-loop security system circuit uses devices that are normally what when there is no abnormal condition present?

a. Closed
b. Grounded
c. Open
d. Energized

21. What type of system is typically used in high-security buildings and continuously occupied facilities such as hospitals, universities, airports, and industrial plants?

a. Local alarm system
b. Supervisory system
c. Residential security system
d. Stand-alone access system

22. What type of protection monitors one particular object?

a. Point-of-entry protection
b. Specific area protection
c. Spot protection
d. Perimeter protection

23. What type of protection monitors only one defined location?

a. Specific area protection
b. Spot protection
c. Point-of-entry protection
d. Perimeter protection

24. What type of security protection monitors specific points that might allow entry to or exit from a secure area?

a. Spot protection
b. Specific area protection
c. Point-of-entry protection
d. Volumetric protection

25. What credential technology requires a combination of two or more credentials in a single unit before allowing access?

a. Single-factor credential technology
b. Mixed credential technology
c. Passive credential technology
d. Proximity-only credential technology

26. What may an access control system use to store an access-control database and confirm a match with stored information?

a. A computer
b. A relay module
c. A modem
d. A line-seizure device

27. What process involves collecting information from the user and verifying credentials?

a. Authorization
b. Enrollment
c. Authentication
d. Partitioning

Communications, CCTV & Paging

32. What is the main purpose of installing telephone line-seizure devices upstream of other telephone devices?

a. To increase the voltage available to downstream telephone devices
b. So that, if necessary, an emergency control panel can take control of the telephone line
c. To prevent all outgoing calls from the building
d. To convert analog telephone signals to digital data

34. What is a common type of paging system used in commercial settings?

a. High power-amplified system
b. Low power-amplified system
c. Central amplified system
d. Passive paging loop

Elevators

36. What type of elevator uses a worm-gear reduction unit that allows precise control of elevator position?

a. Hydraulic elevator
b. Gearless traction elevator
c. Geared traction elevator
d. Machine room-less elevator

PLC Troubleshooting & Maintenance

42. What are the two categories of power-supply problems for PLCs?

a. Loss of power and low voltage
b. High frequency and low frequency
c. Ground faults and short circuits
d. Logic faults and communication faults

43. When a control transformer is fully loaded, what condition indicates that it is overloaded?

a. A voltage drop of more than 2%
b. A voltage drop of more than 5%
c. A voltage rise of more than 5%
d. Any measurable voltage drop

44. When troubleshooting PLC hardware, what is the first step in testing a circuit?

a. Force the suspected output
b. Replace the processor
c. Turn power off and apply lockout/tagout (LOTO)
d. Check the program for a faulted rung

45. Forcing a PLC during what process verifies that the wiring to input devices and the action are correct?

a. Documentation
b. Troubleshooting
c. Program backup
d. Battery maintenance

46. What is the first step in troubleshooting a PLC application?

a. Replace the input module
b. Cycle power to the PLC
c. View the program
d. Force all outputs off

47. When an input terminal of a PLC is forced, how does the LED display rectangle for that input terminal appear?

a. It appears solid
b. It appears empty
c. It flashes continuously
d. It displays an X

48. When using PLC programming software, which key generates a help window for the selected dialogue box, instruction, view, or window?

a. F1
b. F2
c. F5
d. Esc

49. When using the TND instruction for PLCs, where does the debugging process typically start?

a. At the last rung of the program
b. At the first output instruction
c. At the beginning of a program
d. At the first forced input

50. What are the small windows in PLC software that allow users to make choices and enter information called?

a. Data tables
b. Dialogue boxes
c. Status files
d. Instruction blocks

55. What does visual PLC maintenance cover?

a. Only the processor and power supply
b. The PLC enclosure and software files only
c. The PLC enclosure, PLC-related components, PLC input devices, and output components
d. Only field input and output wiring

56. What is RUN MODE?

a. A PLC state where the operating cycle is stopped while the CPU is ready to accept information
b. A PLC state where the operating cycle is running continuously to control a machine or process
c. A software state used only to compare an EEPROM with the processor program
d. A monitoring state in which outputs are disabled but inputs continue to scan

57. What is software and program verification?

a. A process that ensures a PLC can be accessed by a PC and that the PC has a current copy of the program in the PLC
b. A process for replacing PLC batteries and documenting the installation date
c. A visual inspection of the enclosure, cooling fans, and terminal strips
d. A procedure used to verify only the incoming three-phase voltage

58. What do instruction comments provide?

a. Information about the entire PLC enclosure
b. Information about a specific instruction associated with a specific address
c. A history of every program download
d. A list of all processor faults and warnings

NEC Selective Coordination & GFPE

59. According to the NEC, what is selective coordination?

a. The localization of an overcurrent condition to restrict outages to the circuit or equipment affected, accomplished by the selection and installation of OCPDs and their ratings or settings
b. The maximum current that an overcurrent protective device can interrupt at rated voltage
c. The use of two overcurrent protective devices in series to increase the available interrupting rating
d. The adjustment of all OCPDs so that upstream and downstream devices open simultaneously

63. Where is ground-fault protection of equipment not permitted or not required under the noted exception?

a. It is not permitted for fire pump circuits and is not required for a continuous industrial process where a non-orderly shutdown would introduce additional or increased hazards
b. It is not permitted for emergency systems and is not required for hospitals
c. It is not permitted for elevators and is not required for data centers
d. It is not permitted for generator feeders and is not required for commercial buildings

64. When do ground-fault protection requirements NOT apply to a service disconnect?

a. When the service disconnect is rated 1,000 amperes or more

b. When the system is a 480/277-volt solidly grounded wye system

c. When it is used for a continuous industrial process where a non-orderly shutdown would introduce additional or increased hazards

d. When the disconnect supplies more than one feeder

65. What are the two common methods used for ground-fault protection of equipment?

a. Differential and inverse-time methods

b. Zero-sequence method and residual method

c. Line-to-neutral and phase-to-phase methods

d. Core-balance and series-trip methods